

spur demand, these workplaces could expand employment. In particular, Generative AI is argued to reshape organizational hierarchies by displacing entry-level positions (Brynjolfsson, Chandar and Chen, 2025). Section 3.3 analyzes these possibilities by estimating the effects of employer chatbot initiatives on workplace employment, early-career jobs, and job churn.

3. *Job mobility and sorting:* If chatbots are especially useful for newcomers to an occupation, adopters may be more likely to switch occupations (Autor, 2024). Furthermore, AI chatbots may prompt workers who want to use them to move into firms that encourage their use. We examine these implications at both the worker and workplace levels.

We use a series of difference-in-differences analyses to identify the relevant effects, comparing adopters and non-adopters before and after the introduction of AI chatbots and examining outcomes at both the worker and workplace levels. Appendix A.2 draws on our theoretical framework to clarify the identifying assumptions and to provide a structural interpretation of the estimates. Here, we highlight the key empirical challenges in our analysis.

A first concern in our analysis is that adoption may remain a relatively weak treatment: workers may struggle to use the tools effectively, and firms may provide limited guidance or infrastructure to support their implementation. Section 1.4 showed that employer encouragement is critical for unlocking both higher use and greater reported benefits. We use these “AI front-runner” workplaces—those with encouraged use—as our primary workplace treatment, as they offer a window into the effects of AI chatbots as employers embrace the tools. We also examine the effects of additional investments in employer-provided chatbots and training. At the individual level, we also analyze heterogeneity by intensity of use and reported gains.

A second key concern is that adopters may have fared differently in the labor market even without AI chatbots. We take two steps to address this issue. First, we control

for workers’ pre-determined characteristics—including gender, age, and labor market experience—to ensure these factors do not drive our estimates (e.g., adopters being younger and naturally on upward earnings trajectories). Second, we leverage our panel data to implement a difference-in-differences approach indexed to November 2022, the release date of ChatGPT. This allows us to control for time-invariant differences between workers (e.g., high-ability adopters who would have earned more regardless) and examine whether adopters experienced differential changes after this date. Additionally, we assess whether adopters were on distinct labor market trends even before AI chatbots became available, enabling us to control for potential pre-existing differences.

A third question is timing: When should we date the arrival of AI chatbots, and when should we expect their effects on labor market outcomes to appear? We use the launch of ChatGPT in November 2022 as a before-and-after moment in an event-study design, marking a sharp rise in public awareness of AI chatbots.⁸ We also separately examine effects for early adopters—workers observed using chatbots in our 2023 survey round. Still, even if adoption occurs rapidly, labor market effects may take time to materialize. We examine these adjustment dynamics in two ways. First, we implement a dynamic difference-in-differences design to trace how effects unfold monthly following the introduction of AI chatbots. Second, we analyze heterogeneity across occupations that vary in the flexibility of labor rules.

A final concern is that difference-in-differences identifies only *differential* effects for adopters—the “missing intercept.” Appendix F explores potential spillovers to non-adopters from several angles, all of which suggest that AI chatbots have had no meaningful impact on their labor market outcomes to date.

⁸This timing is motivated by the rapid uptake of ChatGPT among workers: most adopters began using it within a year of its launch (Humlum and Vestergaard, 2025). Appendix E.1.3 shows that ChatGPT remains the dominant AI chatbot to date. Moreover, event studies centered on its launch in November 2022 have become common in the literature (e.g., Brynjolfsson, Chandar and Chen (2025); Eisfeldt et al. (2024); Lichtinger and Hosseini Maasoum (2025); Schubert (2025); Teutloff et al. (2025)), facilitating comparison across studies.

2.1 Regression Specifications

Let Y_{it} denote a labor market outcome (e.g., earnings) for worker i in month-year t , let X_i represent a vector of workers' pre-determined characteristics (age, gender, experience, and occupation FEs), and let A_i indicate whether the worker has adopted AI chatbots. We examine both worker-level effects (e.g., worker earnings by chatbot usage) and workplace-level effects (e.g., workplace employment by chatbot initiatives).

Dynamic Difference-in-Differences. To assess overall impacts, we employ a dynamic difference-in-differences specification that compares the monthly outcomes of adopters and non-adopters before and after the introduction of AI chatbots:

$$Y_{it} = \underbrace{\sum_{\tau} \lambda_{1\tau} X_i \mathbf{1}_{\{t=\tau\}}}_{\text{Pre-Determined Controls}} + \underbrace{\sum_m \lambda_{2m} \mathbf{1}_{\{m=m(t)\}}}_{\text{Seasonality}} + \underbrace{\sum_{\tau \neq 2022M11} \beta_{\tau} A_i \mathbf{1}_{\{t=\tau\}}}_{\text{Dynamic Diff-in-Diffs}} + \beta_0 A_i + \varepsilon_{it}, \quad (1)$$

where λ_{2m} are month fixed effects that control for seasonality in labor market outcomes. The parameters of interest, β_{τ} , capture the differential changes in labor market outcomes for adopters, indexed to November 2022, the release of ChatGPT. We estimate Equation (1) by OLS and cluster the standard errors by workplace-occupation level.

Pooled Difference-in-Differences. To examine heterogeneous impacts across multiple dimensions, we use a pooled specification that compares average outcomes of adopters and non-adopters before and after the introduction of AI chatbots. To capture dynamics, we estimate separate effects for each year following their introduction.⁹ Our estimating equation reads:

$$Y_{it} = \underbrace{\sum_{\tau} \lambda_{1\tau} X_i \mathbf{1}_{\{t=\tau\}}}_{\text{Pre-Determined Controls}} + \underbrace{\sum_m \lambda_{2m} \mathbf{1}_{\{m=m(t)\}}}_{\text{Seasonality}} + \underbrace{\lambda_3 t + \lambda_4 A_i t}_{\text{Time Trends}} + \underbrace{\sum_{y=2023}^{2024} \beta_y A_i \mathbf{1}_{\{y(t)=y\}}}_{\text{Pooled Diff-in-Diffs}} + \beta_0 A_i + \varepsilon_{it}, \quad (2)$$

⁹To align with our dynamic difference-in-differences specification (Equation (1)), which defines post-periods as months after 2022M11, we include 2022M12 in the 2023 estimate.

where we add adopter-specific time trends to ensure our pooled difference-in-differences estimates are not driven by spurious trends in outcomes. The parameters of interest, β_y , measure how labor market outcomes for adopters changed in year y after the introduction of AI chatbots, compared to the period before their arrival. We estimate Equation (2) by OLS and cluster the standard errors by workplace-occupation level.

3 Results

3.1 Worker Earnings

Figure 2, Panels (a) and (b) examine earnings of chatbot adopters, distinguishing between those with and without employer encouragement. Each group is compared to non-encouraged non-adopters—workers who are most insulated from AI chatbots, either through their own non-use or that of their coworkers.

Panel (a) begins by presenting differences in means, controlling for workers’ pre-determined characteristics—occupation, gender, age, and experience. Adopters appear to earn substantial premia of about 9% for those with employer-encouraged usage and 4% for those without. However, leveraging the panel dimension of our administrative data reveals that these earnings differentials entirely predate the arrival of AI chatbots. When indexing differences to November 2022, the difference-in-differences estimates in Panel (b) show no differential changes for adopters following the introduction of AI chatbots. The estimates remain flat even among adopters whose employers explicitly encourage chatbot use. Appendix E.2.1 shows that these null results hold separately for workers’ hourly wages, intensive-margin hours, and extensive-margin employment.¹⁰

Figure 2, Panel (b) provides three key insights. First, two years after their launch, AI chatbots have had minimal earnings impact on workers who use them. The confidence intervals of our dynamic difference-in-differences estimates rule out changes larger than

¹⁰The results for adopter earnings echo the pattern documented by Krueger (1993) and critiqued by DiNardo and Pischke (1997): although chatbot use is correlated with sizable cross-sectional earnings premia, these differentials vanish once accounting for positive selection.